Why impact also needs energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Explain why momentum does not tell the whole impact story.

SUCCESS CRITERIA - I can...

1. Compare momentum and kinetic energy as different quantities.
2. Identify speed as a squared factor in kinetic energy.
3. Name where energy can go in an impact.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “momentum” and “kinetic energy” to explain your model or result. Sentence starters are optional.

momentum: mass times velocity, including direction

kinetic energy: energy associated with motion

scalar: a quantity described by size without a vector direction

01 Warm-up: bring back an earlier idea

An impactor has 3 times the original mass at the same speed and stopping time.
Compare momentum and average stopping-force size.

Sentence starter: Both ... . They differ in ... because ... .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: $p = mv$; $K = \frac{1}{2}mv^2$.
Momentum includes direction; kinetic energy is a scalar. Neither alone gives a complete damage forecast.

Worked reasoning: A 2 kg cart at 3 m/s has momentum 6 kg·m/s and kinetic energy $\frac{1}{2}\times2\times3^2=9$ J. A 1 kg cart at 6 m/s has the same momentum but 18 J of kinetic energy.

Your turn: Draw labeled momentum arrows and energy bars for the two carts.

Sentence starter: The ... in my model represents ... . Its label shows ... .



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Two equal-mass carts move at speeds 2 and 4 m/s.
Predict how their momenta and kinetic energies compare before calculating.

Sentence starter: Both … . They differ in … because … .



Same mass; 2 m/s Same mass; 4 m/s

Marzano check - circle: 1 2 3

04 Apply the physics

The lab carts in question 3 each have mass 2 kg. One moves at 2 m/s and the other at 4 m/s.
Calculate both kinetic energies.
Draw bars to test your prediction about the effect of doubling speed.

Sentence starter: I predict … . My model supports this because … .

Initial **Transfers** **Final**



Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

05 Explain your reasoning

List three forms or destinations of energy after a deforming collision. Does “less kinetic energy” mean “energy destroyed”?

Sentence starter: I notice … . This supports … because … .

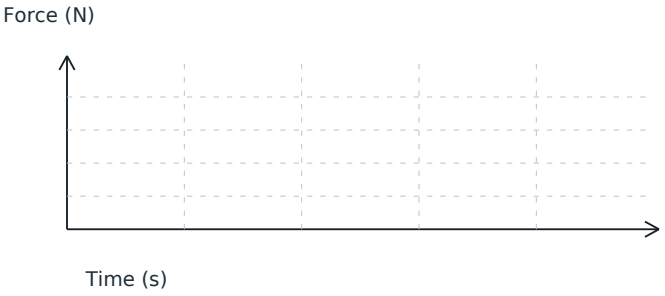
Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A 24 N net force acts right for 3 s on a 4 kg cart initially at rest.
Find impulse and final velocity.
Show the signed impulse on the force-time model.

Word help: impulse = net force acting over time; change in momentum

Sentence starter: I found … [unit]. This value means … in this situation.



Marzano check - circle: 1 2 3

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Continue down each column.

Sentence starters are optional. You may plan in any language.

07 Exit check

For the same mass, speed triples. By what factor does kinetic energy change?
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3

08 STRETCH: test the model

Find two carts with equal mass but speeds in a 1: 2 ratio.
Compare momentum and energy multipliers.
Use the energy model to show the quantities you compare; label units and any unknowns.

Sentence starter: Both … . They differ in … because … .

Initial **Transfers** **Final**

Draw bars or transfer arrows; label forms and joules.

Marzano check - circle: 1 2 3